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CON E 652
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In-Class Project – Part 1 & Homework 5

1. Sensors, Application, and Target Activities

Sensor Selection

This experiment uses 6-axis IMU data collection from an Apple Watch. Tri-axial accelerometer and gyroscope signals are obtained through Apple's CMDeviceMotion sensor-fusion pipeline, which additionally provides attitude estimates (yaw, roll, pitch), a gravity vector, and a calibrated magnetic field, though not all of these features will be used in the classification experiment. Heart rate (BPM via Apple's HealthKit) is recorded as a supplementary feature, and its impact on classification accuracy will be evaluated in early stages of the experiment to determine its utility. The magnetometer data may be excluded because ferromagnetic interference from gym equipment (dumbbells, weight machines) are known to make magnetometer reading unreliable in the target environment.

Data Recording Application

SensorLog (v6.1, developed by Dr. Bernd Thomas) is a research-grade iOS/watchOS sensor-logging application that has been used in published academic research. It supports simultaneous multi-sensor logging at up to 50 Hz on the Apple Watch, exports data as CSV files with descriptive column headers (e.g. motionUserAccelerationX(G), motionRotationRateY(rad/s)) and includes a built-in integer labeling feature for real-time data annotation during recording. The CSV export and HTTP streaming formats make integration with Python/pandas workflows trivial.

Target Activities

The experiment targets gym exercise recognition across 5 categories: (1) upper-body resistance exercises such as dumbbell bicep curls, dips, and pull-ups; (2) cardio activities including walking and running; (3) lower-body exercises such as squats; (4) core exercises performed using an "ab roller" device; (5) rest/transition periods. This set spans a range of motion profiles from highly periodic gait patterns (walking, running) to rhythmic but slower resistance movements (curls, presses) and constrained wrist motion (core work), providing a meaningful multi-class classification challenge.

2. Sampling Frequency, Nyquist Frequency, and Filtering

Sampling Frequency

SensorLog will be configured to record all sensors simultaneously at 50 Hz. This rate was chosen for several reasons: it matches the UCI HAR benchmark standard, aligns with Apple's own WWDC recommendations for workout analysis, and keeps file sizes and battery consumption manageable. Although SensorLog supports up to 100 Hz for a single sensor, 50 Hz is the maximum when logging all sensors simultaneously on the Apple Watch. Additionally, a 2024 work published in IEEE J Biomed Health Information named "Seven Things to Know about Exercise Classification with Inertial Sensing Wearables" found that reducing sampling frequency from 100 Hz down to approximately 20 Hz resulted in less than a 3% difference in classification accuracy, and that models trained on data that were downsampled to 50 Hz performed as well as models trained at 100 Hz.

Nyquist Frequency and Aliasing

By the Nyquist-Shannon sampling theorem, a 50 Hz sampling rate can faithfully represent signal components up to 25 Hz. Human body motion during gym exercises is predominantly below 10 Hz, with gait frequencies falling around 1-3 Hz and repetitive lifting motions around 0.3-1.0 Hz, so a 25 Hz Nyquist limit is well above the frequencies of interest. The risk of aliasing is minimal for the targeted activities.

Filtering Strategy

Despite the low aliasing risk, a low-pass filter will be applied to remove high-frequency sensor noise (electrical noise, mechanical vibration) that lies above the band of interest. Following the UCI HAR benchmark configuration, a low-pass filter with a 20 Hz cutoff will be applied using scipy.signal.filtfilt() (yes, that's actually the name of the function) to preserve all exercise-related motion content while attenuating noise.

3. Number of Runs and Data Segmentation

Dataset Size Targets

The target sample size is 200 labeled windows per exercise class, which may be limited by available time to collect data. With 3-second windows at 50% overlap, each 10-rep set of a resistance exercise (lasting approximately 30 seconds) yields roughly 15-18 windows. Three sets per session produce approximately 50 windows per exercise per session. Therefore, approximately 4 recording sessions per exercise are needed to reach adequate training volume. For a dataset covering a maximum of 6 exercise classes (plus rest), this translates to 4 training sessions (or 2 gnarly sessions) collected across multiple days to capture all required data.

Segmentation Strategy

The recoded continuous sensor streams will be segmented into fixed-length analysis frames using a sliding window approach. Windows of 3 seconds at 50 Hz yield 150 samples per window. A 50% overlap between consecutive windows will be applied, which doubles the effective number of training samples and ensures that no exercise repetition straddles a window boundary without being fully captured in at least one window. The 3-second window length is chosen because it captures one full repetition of most target exercises.

4. Sensor Placement and Securing the Device

Placement

The Apple Watch will be worn on the non-dominant wrist (left wrist in my case), which is a standard position for this type of device. A wrist-worn sensor is well-suited for this experiment because upper-body exercises produce strong, distinctive signals at the wrist, and cardio exercises like walking and running generate clear periodic accelerometer patterns through the arm and body. Core exercises present the greatest challenge for wrist-worn sensors, as wrist motion is more or less fully constrained; however, the gravity vector captured though CMDeviceMotion (reflecting torso angle changes transmitted through the wrist) and heart rate provide supplementary discriminating features.

Securing the Device

Consistency in band tightness is the critical factor , so to avoid motion artifacts from the watch shifting or rotating on the wrist, the watch band will be tightened snugly before each session so that the watch body sits firmly against the top of the wrist with no lateral play.

5. Recording Duration, Window Length, and Number of Sessions

Recording Duration per Activity

Each exercise within a session consists of 3 sets of 8-12 repetitions, with 30-60 seconds of rest between periodic sets. A single set lasts approximately 30 seconds (or more) depending on the exercise and the tempo (and how much coffee I've had at the time the data is collected). Including rest intervals and transition time, each exercise takes roughly 4-6 minutes of total recording time per session. A full recording session covering 3-5 exercises runs approximately 15-30 minutes, well within SensorLog's constraint of reliable multi-sensor background logging for sessions under 1 hour.

Window Length

A 3-second window (150 samples at 50 Hz) with 50% overlap (1.5 seconds per step) is proposed. This duration is chosen because it is long enough to capture one full repetition of most exercises while remaining short enough for responsive classification. A 3-second window at 50 Hz produces 150 time-series samples per channel, which is a manageable size for both feature extraction and deep learning stages of the experiment.

Number of Sessions

A total of 4-6 30-minute recording sessions will be collected over the period of several days. Spreading collection across different days is beneficial because between-day variability in form, fatigue level, grip tightness, and watch position is a primary source of real-world generalization challenge. A model trained on data from a single session could overfit to the specific conditions of that day.

6. Label Assignment and Reducing Label Noise

Real-Time Labeling

SensorLog's builti-in integer labeling feature will be used to assign coarse real-time labels during recording. Each exercise is assigned a unique integer code (e.g. 0 = rest/transition, 1 = bicep_curl, 2 = dip, 3 = pull-up, 4 = lateral _raise, 5 = walk/run). The label is changed at the start or end of a set by tapping a label button on the SensorLog app on the Apple Watch. Including 2-3 seconds of standing rest before and after each set will be employed to create clean transition boundaries that will simplify post-collection label refinement.

Post-Collection Refinement

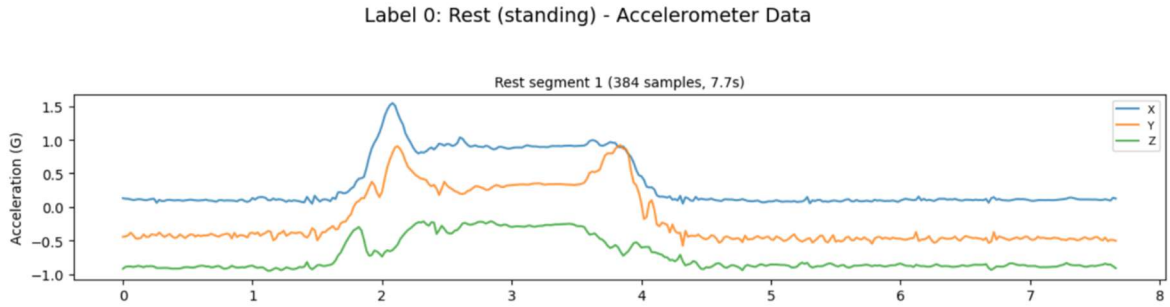
SensorLog's labeling feature will be the main recourse for handling labeling noise, but any labeling issues identified at time of collection can be refined using pandas or targeted re-collection if labels are incorrectly applied by the feature. Published research indicates that classifiers can tolerate approximately 15-20% of label noise, so even imperfect real-time labels work well when combined with post-collection review.

7. Pilot Test and Preliminary Visualizations

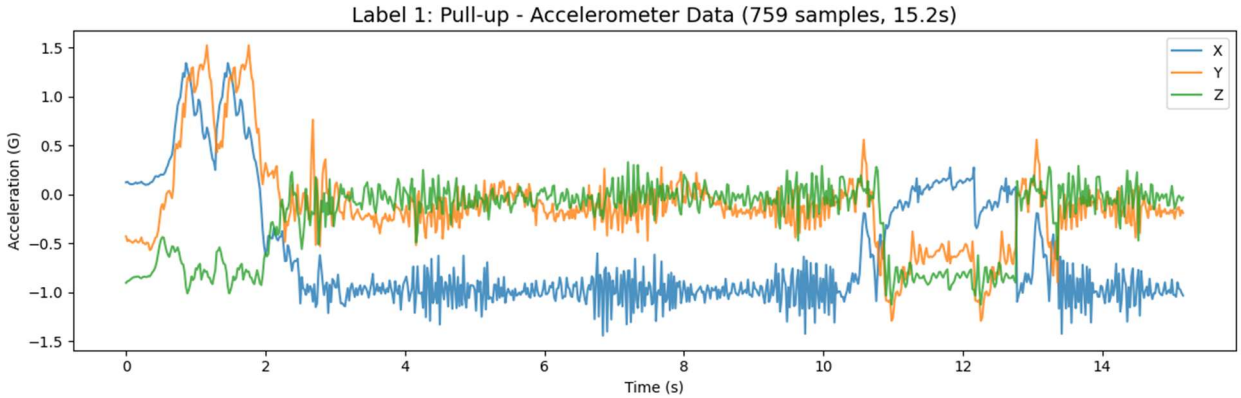
A pilot test was conducted to verify the recording setup, data export pipeline, and signal quality. Short recordings (about 3-5 seconds each) were captured for four representative activities: rest (standing), pull-ups, dips, and bicep curls. SensorLog's real-time labeling feature was used in the test, and the single CSV file was exported into Python from the

iPhone to a Windows machine via iCloud Drive, and the following visualizations for each label were produced:

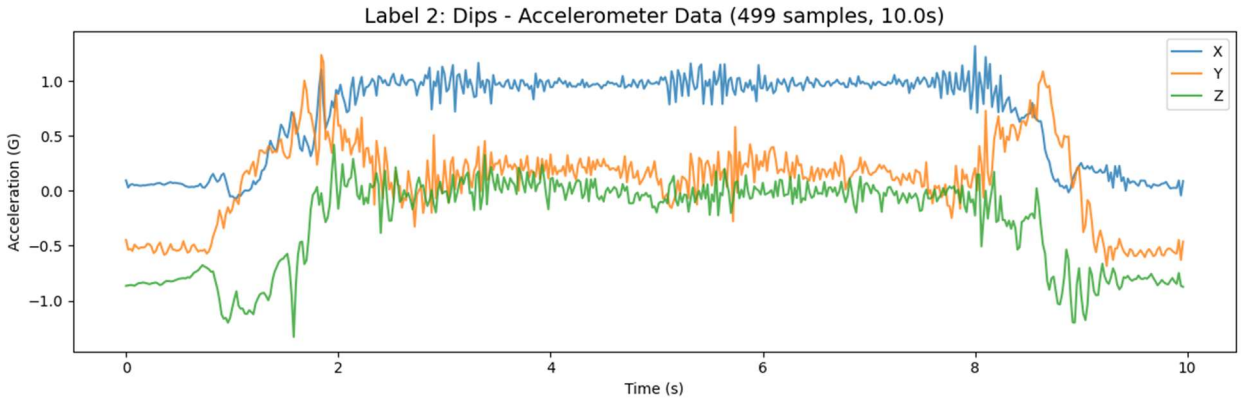
Label: 0 – rest (standing)



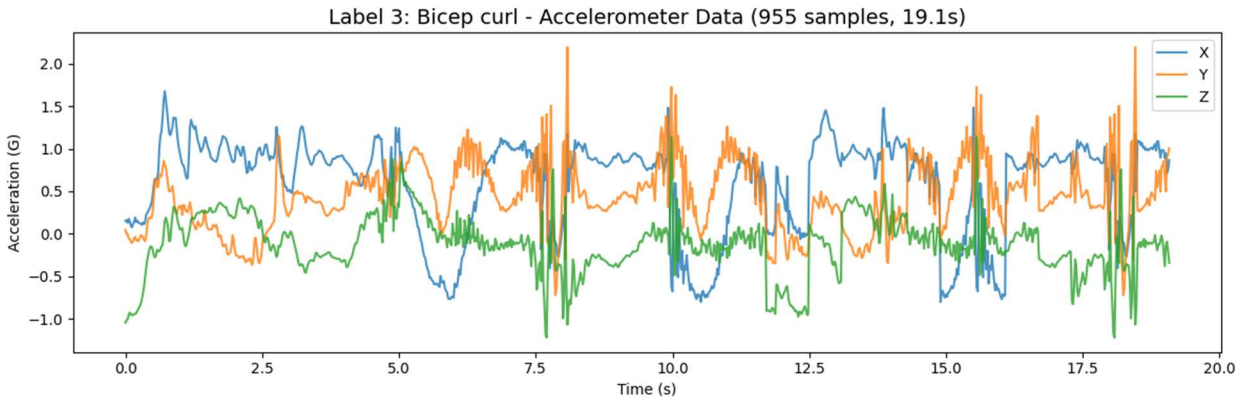
Label: 1 – pull-ups



Label: 2 – dips



Label: 3 – bicep-curls



8. Proposed Evaluation Metrics

For multi-class activity recognition, overall accuracy alone is not sufficient because it can be misleading when class frequencies are imbalanced (e.g. rest/transition segments may be over-represented), therefore the following metrics are proposed:

Confusion Matrix

The confusion matrix is the foundational multi-class evaluation tool, showing for each trie activity class how many windows were classified into each predicted class. This immediately reveals which specific activity pairs are being confused and guides targeted improvements to feature engineering or class definitions.

Precision and Recall (Per-Class)

Precision measures the fraction of windows predicted as a given class that truly belong to that class (i.e., the reliability of positive predictions). Recall measures the fraction of true windows of a given class that were correctly identified (i.e., completeness of detection). These per-class metrics are especially important for exercises that may be difficult to distinguish, such as core exercises where the wrist-worn sensor signal is weak.

F1-Score (Macro-Averaged)

The F1-score is the harmonic mean of precision and recall and provides a single balanced measure for each class. Macro-averaging the F1-score across all classes gives equal weight to each activity regardless of its frequency in the dataset. This is the primary single-number metric for model comparison, as it penalizes classifiers that perform well on common classes but poorly on rare ones.

Balanced Accuracy

Balanced accuracy averages the recall across all classes, accounting for class imbalances without requiring explicit resampling. If the rest/transition class is substantially larger than the exercise classes, balanced accuracy will give a more realistic assessment of model performance than standard accuracy.

Cross-Validation Accuracy

All metrics will be computed using leave-one-session-out (LOSO, because I'm cool) cross-validation: the model is trained on all recording sessions except one, tested on the held-out session, and this is repeated for every session. LOSO prevents temporal information leakage that would occur with random k-fold splitting on time-series data, and it provides a realistic estimate of how the model will perform on unseen future sessions. Published research indicates that random k-fold can inflate accuracy metrics by 5-15% compared to LOSO on HAR tasks.

Summary of Proposed Metrics

Metric	**Purpose**
Confusion Matrix	Identify which activity pairs are confused with each other
Precision (per-class)	Reliability of each class prediction
Recall (per-class)	Completeness of detection for each activity
F1-Score (macro)	Primary balanced single-number metric for model comparison
Balanced Accuracy	Class-imbalance-aware overall accuracy
LOSO CV	Prevent temporal leakage; realistic generalization estimate